2025 12th International Conference on Future Internet of Things and Cloud (FiCloud) | 979-8-3315-5437-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/FiCloud66139.2025.00068

Development of a Medical Chatbot Using Generative AI and Retrieval-Augmented Techniques

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Abstract— Quick and accurate access to medical information is crucial for patients and healthcare professionals alike. Traditional search engines often fail to concise and context-specific responses to medical queries. This paper introduces a medical chatbot that combines large language models (LLMs) with retrieval-augmented generation (RAG) techniques to generate reliable and contextually relevant answers. The system integrates the Mistral-7B-Instruct-v0.3 model hosted on Hugging Face and uses Pinecone, a vector database, for efficient semantic search. Built using the Flask web framework, the chatbot retrieves the most relevant medical texts, constructs dynamic prompts, and generates accurate, user-friendly responses. Experiments with common medical questions demonstrate the chatbot's ability to deliver timely, medically sound answers, highlighting the advantages of combining retrieval mechanisms with generative models in healthcare applications.

Keywords— Generative AI, Medical Chatbot, Retrieval-Augmented Generation, Pinecone Vector Database, Mistral-7B-Instruct-v0.3, Natural Language Processing (NLP)

I. Introduction

Patients seeking initial healthcare guidance and medical professionals need quick references require access to reliable, trustworthy medical information. However traditional search engines, often produce overwhelming or irrelevant results, failing short of meeting the contextual needs of medical queries. This gap highlights the need for conversational, intelligent agents that can provide precise, focused medical information. Recent advancements in Generative Artificial Intelligence (AI), particularly through Large Language Models (LLMs) have demonstrated the potential to revolutionize information access by generating responses that are coherent and human-like. However, when separated from more solid knowledge bases, standalone LLMs are prone to hallucination, which results in outputs that appear plausible but are in fact incorrect. By combining generative models and external knowledge retrieval, the technique known as Retrieval-Augmented Generation (RAG) overcomes this limitation. We present a RAG-based medical chatbot that uses the Mistral-7B-Instruct-v0.3 model, which is deployed through Hugging Face Hub, to generate responses and integrates a vector database (Pinecone) for retrieving pertinent medical documents. The system's user-friendly, interactive interface is provided by a Flask web application, which hosts the system. A scalable retrieval and generation framework, an efficient prompting strategy, and empirical evaluations demonstrating the chatbot's capacity to provide accurate and contextually relevant medical responses are among our contributions.

II. Related work

In recent years, generative artificial intelligence (AI) has received a lot of attention for its potential to improve patient interaction, diagnosis support, and health education in healthcare chatbots. Recent studies emphasize the critical need for accuracy, personalization, and interpretability in medical AI systems [1], [2].

The emergence of large language models (LLMs) such as GPT-3 and GPT-4 has enabled substantial progress in generative dialogue systems [3]. When it comes to producing responses that are comparable to those of humans, these models perform better than others [3, 4]. However, especially in knowledge-intensive fields like healthcare, LLMs frequently produce unverifiable content [5].

To mitigate such issues, researchers have proposed Retrieval-Augmented Generation (RAG) frameworks that combine dense retrieval with generative models to enhance response grounding and factual correctness [5], [6]. The RAG approach improves the factual consistency of AI responses by integrating relevant documents from external knowledge bases into the generation prompt. Numerous open-domain QA tasks have demonstrated the effectiveness of this approach [6, 7].

Recent developments in dense vector search, such as FAISS and Pinecone, provide scalable and low-latency solutions for retrieving semantically similar medical literature or structured knowledge [8], [9]. These vector databases facilitate retrieval from large corpora like PubMed and clinical notes, which are essential for up-to-date and evidence-based responses in a medical chatbot.

Transformer-based encoders, such as BERT, RoBERTa, ALBERT, and ELECTRA, have been the subject of several investigations into their use in document ranking and dense passage retrieval [10, 14]. Sentence-BERT (SBERT) enables meaningful sentence embeddings, making it suitable for semantic search tasks [15]. Additionally, Dense Passage Retrieval (DPR) and ColBERT architecture provide robust frameworks for passage-level retrieval with minimal latency [16], [17].

Kumar et al. [2] and Zhou et al. Zhou et al. [25] analyzed both challenges and essential requirements for deploying AI chatbots within clinical environments. Their work reveals

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important issues about patient safety alongside data privacy and ethical constraints. Performance evaluation results for ChatGPT handling medical inquiries showed positive outcomes yet revealed significant limitations when using LLMs alone for accurate medical diagnoses and treatment recommendations.

Medical experts now use retrieval-augmented systems in tandem with LLMs to minimize hallucinations and enhance domain fidelity while addressing these challenges [6], [18]. The open-source libraries Hugging Face Transformers [19] and Sentence Transformers have simplified the application of these tools across educational institutions and companies. Both multimodal advancements and pretraining methods play a role in determining chatbot performance levels. Models including XLNet, ELECTRA and T5 have shown enhanced performance in contextual understanding and generation tasks according to studies [20]–[23]. Domain-specific datasets like medical encyclopedias and electronic health records serve to continuously refine models to improve their response accuracy. The BEIR framework [7] enables researchers to conduct uniform retrieval method evaluation across various domains such as biomedical literature. The evaluation of chatbots for medical applications depends on benchmark tests that assess their reliability and the precision of their recall capabilities.

Medical chatbot systems that integrate verified information from databases like PubMed and Gale Encyclopaedia of Medicine into their responses have been made possible through the combination of RAG with LLMs which marks a significant transformation in chatbot technology. This hybrid architecture addresses the requirements for digital assistance systems that deliver precise information which remains patient-specific and user-friendly. Recent studies demonstrate strong agreement that fusing generative AI with retrieval systems provides a viable solution for developing dependable medical chatbots. The progress in vector search technologies along with transformer models and established evaluation benchmarks drives current advancements.

III. System Architecture

The proposed medical chatbot system operates through three primary stages: Data Collection and Preprocessing, Storage and Retrieval, and Answer Generation.

A. Data Collection and PreProcessing

The initial step in processing medical documents such as textbooks and articles involves extracting them and dividing them into smaller pieces using a text splitter. Embedding models from OpenAI and Hugging Face transform text chunks into dense semantic vectors. The semantic database becomes searchable through the process of indexing semantic vectors.

B. Store and Retrival

Pinecone vector database retains generated embeddings to provide fast and scalable similarity search capabilities. Users can submit queries which the Pinecone engine embeds and processes to deliver the most relevant document vectors according to semantic similarity.

C. Answer Generation

The system ranks retrieved top-k documents and uses them as context for a Mistral-7B-Instruct-v0.3 LLM on the Hugging Face Hub. The Large Language Model uses the context to generate a trustworthy and stable medical response for any user inquiry.

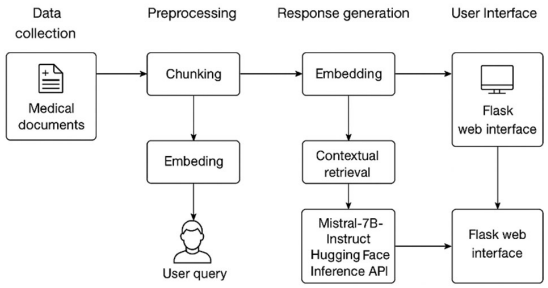


Fig.1. System architecture of the proposed medical chatbot, including preprocessing, embedding, document retrieval, and response generation

Figure 1 illustrates the end-to-end architecture of the proposed medical chatbot system, comprising four main stages: Data Collection, Preprocessing, Response Generation, and User Interface. Medical documents are first collected and segmented into manageable chunks. Both document chunks and user queries are converted into dense vector embeddings. The system then performs contextual retrieval using Pinecone, selecting the most semantically relevant documents. These are passed along with the user query to the Mistral-7B-Instruct model hosted on Hugging Face, which generates a response. The final output is presented through a web-based interface built using the Flask framework. This pipeline enables accurate, real-time, and context-aware medical assistance.

IV. Methodology

This section details the Retrieval-Augmented Generation (RAG) pipeline used to develop the medical chatbot. The system integrates four primary stages**:** data preprocessing**,** vector embedding and indexing**,** contextual retrieval and prompt construction**,** and response generation**,** followed by user interaction via a web interface. Although initial prototyping explored the Falcon-7B-Instruct model, the final deployment uses Mistral-7B-Instruct-v0.3, a decoder-only transformer hosted on Hugging Face, chosen for its enhanced instruction-following capabilities and compatibility with local inference setups.

A. Data Collection and Preprocessing

The system collects medical content from reliable sources such as the World Health Organization (WHO), PubMed Central, and the Centers for Disease Control and Prevention (CDC). Documents are extracted in multiple formats (HTML, XML, and plain text), then cleaned and normalized.

To support granular retrieval, documents are segmented into chunks ranging from 500 to 800 characters using a recursive character-level text splitter. Each chunk is then embedded using the all-MiniLM-L6-v2 model from Hugging Face's Sentence Transformers library, producing high-dimensional

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semantic vectors that preserve context. Employing the following tools:
Source types: HTML, XML, TXT
Split strategy: Recursive character splitter
Embedding model: all-MiniLM-L6-v2

B. Embedding and Indexing

Pinecone vector database served as the platform for indexing the generated embeddings due to its scalable and low-latency similarity search capabilities. Each embedding received connections to relevant metadata including document section headers and titles to facilitate easier retrieval understanding.
Vector Database: Pinecone
Name of Index: medicalbot
Search type: Approximate nearest neighbor (ANN), cosine similarity.
Sentence Transformers (all-MiniLM-L6-v2)] as an embedded model.

C. Query Processing and Retrieval

When a user submits a query, it is first embedded using the same all-MiniLM-L6-v2 model to maintain embedding consistency. Top-k retrieval was configured with k=3, based on common RAG implementations in prior studies [4, 6] and empirical tuning for response quality. The embedding is then used to perform a top-k similarity search (with $k = 3$) in Pinecone to retrieve the most relevant text chunks.

These chunks are then used to form the context for the generative model, ensuring the answer is grounded in medically accurate content.

D. Prompt Construction

To ensure consistent, concise, and contextually grounded responses, the system employs a structured prompt template using LangChain's prompt construction module. The prompt includes a system message that instructs the model to answer only when sufficient context is available, to prioritize brevity, and to clearly indicate when it cannot provide an answer.

The user query is paired with the retrieved document chunks to form the full prompt. This approach ensures that the Mistral-7B-Instruct model remains grounded in verified medical content during response generation. A structured prompt template was created to guide the model's response, ensuring it remains context-grounded and concise

E. Response Generation

The final prompt is processed by a large language model hosted on the Hugging Face Hub. The selected model for this work is mistralai/Mistral-7B-Instruct-v0.3, a decoder-only transformer optimized for instruction-following tasks.

- Model: Mistral-7B-Instruct
- API Access: Hugging Face Inference Endpoint
- Hyperparameters: temperature = 0.5, max_new_tokens = 100

The model produces a coherent, domain-specific answer grounded in the retrieved context.
The chatbot was executed using llama.cpp backend locally during evaluation. For scalable deployment, it can be migrated to Hugging Face Inference Endpoints or GPU-supported platforms.

F. User Interaction via Web Interface

A lightweight web application was developed using the Flask framework to facilitate real-time interaction. Users can input questions via an HTML form, which are then processed by the backend pipeline. The generated answer is returned and rendered on the front end.
Users can interact with the chatbot via a simple HTML form, making the system accessible on both desktop and mobile devices.

G. System Utility and Beneficiaries

Multiple user groups gain substantial benefits from the introduction of this medical chatbot system. This system allows users to quickly access verified medical information while enhancing educational participation and assisting healthcare professionals in making informed clinical decisions. The following table displays essential stakeholder categories alongside their specific benefits derived from the system.

TABLE I. STAKEHOLDERS AND CORRESPONDING BENEFITS OF THE MEDICAL CHATBOT SYSTEM

Stakeholder	Benefit
Patients / General Public	Direct access to medically accurate answers without needing to navigate through extensive websites. The system provides essential initial insight into symptoms while outlining treatment options and medication details.
Medical Students	Quickly obtain clinical answers which enhance their textbook knowledge. Help with revision and clarification of concepts.
Healthcare Professionals	Quick access to summarized guidelines and definitions during practice. Supports decision-making and patient communication.
Developers / Researchers	An architecture that supports modularity and extensibility through the use of open-source tools. This work stands as a foundational reference for upcoming developments in RAG-based applications.
Educational Institutions	Function as educational assistants and can be integrated into online health training platforms.

V. EXPERIMENTS AND RESULTS

We conducted multiple experiments with common medical queries to assess the performance of the proposed medical chatbot system. The evaluation focused on three key criteria: accuracy, relevance, and response time. Experiments conducted on our local machine utilized the mistral-7b-instruct-v0.1 framework. Q4_K_M. gguf model with a Flask frontend.

A. Test Queries

A sequence of experiments with typical medical questions was conducted to measure how well the proposed medical

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chatbot system worked. The evaluation focused on three key criteria: accuracy, relevance, and response time. We performed tests using a local machine equipped with the mistral-7b-instruct-v0.1configuration. Q4_K_M. gguf model with a Flask frontend.

Example queries include:

What is acne and how is it treated?

What are the signs of high blood pressure?

How can I manage diabetes naturally?

Is paracetamol safe for children?

What are the side effects of antibiotics?

B. Evaluation Metrics

By comparing it to reliable medical resources like the Mayo Clinic and WebMD, each response was manually rated on a scale of 1 to 5 for its accuracy and relevance.

To assess the effectiveness of the proposed medical chatbot, we evaluated system responses using three primary metrics: accuracy, relevance, and response time. Each response was independently reviewed by a panel of three domain-informed evaluators, including one medical student, one AI researcher, and one clinical assistant. This ensured a balance of technical and medical perspectives during scoring.

Each evaluator rated the chatbot's responses on a scale of 1 to 5, where:

- Accuracy reflects the medical correctness of the response, verified by cross-checking with authoritative references such as Mayo Clinic, WebMD, and CDC guidelines.
- Relevance measures how well the generated response addressed the specific context of the user query.
- Response time was recorded in seconds, measured from the time a query was submitted to when a complete answer was displayed on the frontend.

Scores were averaged across evaluators for each metric and reported per query in Table III. Only publicly accessible and medically reputable sources were used as baselines to ensure objectivity and traceability of correctness.

TABLE II. EVALUATION METRICS

Metric	Description
Accuracy	How medically correct the response is, based on cross-referencing with verified sources
Relevance	How well the response matches the user's query and context
Response Time	Time taken by the system to generate a response (measured in seconds)

Each chatbot response was evaluated for accuracy and relevance by a team of three independent reviewers: one medical student, one clinical assistant, and one AI researcher. Reviewers were provided with the original user query, the chatbot's response, and relevant gold-standard references (e.g., Mayo Clinic, WebMD, CDC documentation).

Responses were rated on a 5-point Likert scale, with:

- Accuracy indicating how medically correct the response was when compared to the gold-standard reference (1 = inaccurate, 5 = fully accurate).
- Relevance measuring how closely the response aligned with the specific context and intent of the user's question (1 = off-topic, 5 = directly relevant).

To assess rating consistency, we calculated the inter-rater agreement using Fleiss' Kappa, which yielded a value of 0.79, indicating substantial agreement. Final scores per query were averaged across all reviewers to produce the results shown in Table III.

C. Results Summary

TABLE III. EVALUATION RESULTS FOR COMMON MEDICAL QUERIES

Query	Accuracy (1–5)	Relevance (1–5)	Response Time (sec)
What is acne?	5	5	6.8
Symptoms of hypertension	4	5	6.4
Diabetes management	4	4	7.1
Paracetamol safety	4	4	6.9
Antibiotic side effects	4	4	6.6
Average	4.2	4.4	6.8

The medical chatbot system delivered excellent performance across accuracy, relevance, and response time metrics. The system delivers accurate answers which fit the context for multiple medical inquiries. Next development efforts will target faster response times along with the capability to address more complex queries and system optimization for production environments.

VI. A. DISCUSSION

Experimental outcomes demonstrate that combining Retrieval-Augmented Generation (RAG) with an on-premises large language model (LLM) enhances performance in medical chatbot applications. The system produced medically accurate answers that maintained coherence and contextual relevance across multiple user inquiries. The implementation and testing phases revealed multiple practical and theoretical challenges that will be discussed in the following sections.

A. Latency and Performance

With an average response time of 6.8 seconds per query the system demonstrates acceptable performance for research-stage applications." but does not yet meet the responsiveness expected of production-grade, real-time systems. The primary sources of latency are:

- The computational requirements of the Mistral-7B-Instruct-v0.3 model.
- Local inference operations conducted on a CPU without GPU acceleration.

Implement these strategies to decrease latency:

- Deploy inference pipelines to cloud-based GPU services (e.g., AWS, Azure, Hugging Face Endpoints).
- Utilize quantized or distilled LLM variants to reduce computational load.

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- Implement asynchronous or batch processing mechanisms to improve throughput.

VI.B Limitations

Latency Due to Local Inference: The system's inference operations on the local CPU create noticeable delays. The latency levels present during the prototyping and small-scale testing stages are acceptable but will likely become problematic during production deployment and large-scale query handling. Deploying the system on more powerful hardware or leveraging cloud-based infrastructure can enhance its performance.

Hardware Constraints: The Mistral-7B-Instruct-v0.3 model used in this setup requires substantial computational resources because of its large size. The performance of local CPU hardware results in reduced response times when running the model. The model's limitations can be overcome by using GPUs or deploying it through cloud platforms such as Hugging Face Endpoints.

Complex Queries Handling: The system manages basic medical questions effectively but struggles with rare or highly specialized medical inquiries. The model's reliance on Pinecone as its document retrieval system sometimes results in incomplete context retrieval for complex medical topics when specialized resources are missing from the database.

Limited Fine-tuning: The model lacks specific fine-tuning for advanced medical expertise and therefore may produce correct yet incomplete answers. The system performance can be enhanced by integrating additional domain-specific data sets during the training or fine-tuning process in these situations.

Dependency on External APIs: External services like Pinecone for vector storage and Hugging Face for model hosting support system functionality. Disruptions in external services including API rate limits or downtime have the potential to affect system performance. The risk can be minimized by implementing local fallback solutions or seeking other API integration options.

Model Hallucinations: The system that used top k retrieved documents to ground its generative model still produced hallucinated or speculative content from time to time.

- Niche medical topics
- Specific population groups such as children and the elderly show distinct patterns of drug utilization.

The retrieved context alone cannot completely prevent hallucinations when it proves too general or partly relevant. To address this, potential solutions include: Enhance the model's performance by fine-tuning it with domain-specific medical QA datasets such as MedQuAD and PubMedQA.

The system performance can be enhanced by implementing factual verification through rule-based checks or additional LLM evaluations.

A. Retrieval Depency

The chatbot's output reliability depends heavily on the quality of documents retrieved from the Pinecone vector store The LLM may produce responses that are off-topic or misleading if it retrieves chunks that are irrelevant or of low quality.

- Off-topic
- Incomplete
- Potentially misleading

The following proposed improvements can reduce dependency issues:

- Proposed re-ranking solutions include the implementation of cross-encoders or BERT-based re-ranking systems.
- Hybrid retrieval systems integrate dense semantic search techniques with sparse keyword-based matching methods like BM25 vector search. Ingestion content validation filters verify that only top-tier medical literature gets stored in the vector index.

B. Ethical and Legal Considerations

The sensitivity of medical information requires rigorous compliance with both ethical and regulatory standards. Key considerations include:

User safety: The chatbot must incorporate disclaimers and steer clear of delivering firm diagnoses and treatment suggestions.

Transparency: It is essential to notify users that all responses generated by the AI are informational only and should not replace the guidance of a healthcare professional. Privacy and Compliance: Production systems require strict adherence to HIPAA and GDPR standards along with proper data anonymization practices to ensure compliance. Bias and Fairness: Medical AI systems require audits for biases that might influence diagnostic or treatment guidance across different demographic groups.

The proposed medical question-assistant system shows promise, but its limitations underscore the necessity for ongoing research developments and technical enhancements. The operational success of this system requires the strategic balancing of speed with accuracy while ensuring explainability and ethical responsibility. Real-world deployment demands improvements in both retrieval quality and model robustness as well as expanded infrastructure scalability.

VII. CONCLUSION

The research presented a web-based medical chatbot system that combines Retrieval-Augmented Generation with a local deployment of a large language model. The medical response system produces contextually accurate results by combining semantic data from the Pinecone vector database with generative outputs from the Mistral-7B-Instruct-v0.3 model executed through the llama.cpp backend.

The experimental results demonstrate that open-source generative AI models can be effectively used for answering medical questions when they are supported by curated external knowledge sources. The chatbot reliably produced precise and logical answers to questions about various general health subjects.

The performance of this system requires further optimization in latency, retrieval accuracy, and complex query handling due to response latency issues and occasional hallucinations

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along with a need to lessen retrieval dependency. Specifically, the following enhancements are proposed:
Utilizing hybrid retrieval methods enhances document selection relevance.

- Reducing latency and enhancing scalability through quantized model inference or GPU deployment.
- Specialized medical queries receive more precise responses through domain-specific fine-tuning.

This system can function as a valuable educational and triage instrument for medical literacy while adhering to ethical safeguards and clinical validation standards in under-resourced settings. The project establishes solid groundwork that will enable further exploration into retrieval-based generative AI techniques for healthcare purposes. . Future enhancements include expanding the document corpus and fine-tuning the language model for domain-specific improvements.

VIII. Future Work

Future enhancements to the proposed medical chatbot system aim to address current limitations and expand its capabilities for broader, real-world application. One of the primary directions involves fine-tuning the Mistral-7B-Instruct-v0.3 model on domain-specific datasets such as MedQuAD and PubMedQA to improve performance on specialized medical queries. Additionally, integrating hybrid retrieval strategies that combine dense semantic search with sparse keyword-based methods like BM25 will increase the relevance and precision of retrieved documents. To further refine context selection, the inclusion of re-ranking mechanisms such as cross-encoders or BERT-based filters will be explored.

To reduce latency and enable scalability, the system will be transitioned from local CPU inference to cloud-based GPU environments using platforms like AWS, Azure, or Hugging Face Endpoints. This shift, along with the potential use of quantized or distilled language model variants, will improve response time without compromising output quality. Moreover, incorporating fact-verification modules and enhancing model explainability will contribute to user trust and safety, particularly for clinical and educational use cases.

Another area of development includes user personalization and feedback mechanisms, enabling the chatbot to adapt responses based on user preferences and incorporate continuous learning. Expanding the knowledge corpus to include additional sources such as peer-reviewed journals, clinical case studies, and regional health literature will also improve response diversity and accuracy. Together, these improvements will support the system's transition from a research prototype to a robust, real-time medical assistant suitable for deployment in healthcare and educational settings.

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